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BICYCLE FRONT FORKS

Abstract

A front fork comprises a leg including a first tube and a second tube. The first tube has a first end and a second end opposite the first end. The second tube has a third end and a fourth end opposite the third end. The first and second tubes are configured in a telescopic arrangement. The first and second tubes are moveable between a fully extended position and a fully compressed position. The front fork also includes a bumper disposed in the second tube near the fourth end. The bumper is to be contacted during a bottoming-out event. A position of the bumper is adjustable in the second tube.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] This disclosure relates generally to bicycle components and, more specifically, to bicycle front forks with bumpers.

BACKGROUND

[0002] Bicycles are known to have suspension components. Suspension components are used for various applications, such as cushioning impacts, vibrations, or other disturbances imparted to the bicycle during use. A common application for suspension components on bicycles is for cushioning impacts or vibrations experienced by the rider when the bicycle is ridden over bumps, ruts, rocks, potholes, and/or other obstacles. These suspension components include rear and/or front wheel suspension components. For example, some bicycles include a front fork with telescoping legs that incorporate a spring and/or damper system. The front fork compresses and expands when riding over obstacles to help cushion impacts and/or vibrations felt by the rider.

SUMMARY

[0003] An example front fork for a bicycle disclosed herein includes a leg including a first tube and a second tube. The first tube has a first end and a second end opposite the first end, and the second tube has a third end and a fourth end opposite the third end. The first and second tubes are configured in a telescopic arrangement. The first and second tubes are moveable between a fully extended position and a fully compressed position. The example front fork also includes a bumper disposed in the second tube near the fourth end. The bumper is to be contacted during a bottoming-out event before the first and second tubes reach the fully compressed positioned. A position of the bumper is adjustable in the second tube.

[0004] An example front fork for a bicycle disclosed herein includes a leg including an upper tube and a lower tube. The upper tube has a first top end and a first bottom end, and the lower tube has a second top end and a second bottom end. The upper and lower tubes are configured in a telescopic arrangement with the first bottom end of the upper tube disposed in the lower tube. The example front fork also includes a bumper disposed in the lower tube near the second bottom end and an adjuster pin to cause axial movement of the bumper in the lower tube. The bumper is moveable between a lower-most position in which a top of the bumper is a first distance from the second bottom end of the lower tube and an upper-most position in which the top of the bumper is a second distance from the second bottom end of the lower tube, the second distance being greater than the first distance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a side view of an example bicycle that may employ any of the example
[0006] front forks disclosed herein.

[0007] FIG. 2 is perspective view of an example front fork that can be implemented on the example bicycle of FIG. 1.

[0008] FIG. 3 is a cross-sectional view of the example front fork of FIG. 2 taken along line A-A of FIG. 2 and showing an example jounce bumper in one of the legs.

[0009] FIG. 4A is an exploded view of example components of the leg of the example front fork of FIG. 3 including the example jounce bumper.

[0010] FIG. 4B is an exploded view of the same components of FIG. 4A shown in cross-section.

[0011] FIG. 5A is an enlarged view of the callout of FIG. 3 showing the example jounce bumper in a lower-most position.

[0012] FIG. 5B shows the example jounce bumper in an upper-most position.

[0013] FIG. 6A is a cross-sectional view showing the example jounce bumper in the lower-most

position and when an example seal head is making initial contact with the example jounce bumper during a bottoming-out event.

[0014] FIG. 6B shows the example jounce bumper in the lower-most position after being deformed and/or compressed by the example seal head during the bottoming-out event.

[0015] FIG. 7A is a cross-sectional view showing the example jounce bumper in the upper-most position and when an example seal head is making initial contact with the example jounce bumper during a bottoming-out event.

[0016] FIG. 7B shows the example jounce bumper in the upper-most position after being deformed and/or compressed by the example steelhead during the bottoming-out event.

[0017] FIG. 8 is a cross-sectional view of a bottom portion of a leg showing an example jounce bumper used in connection with a spring or damper having an adjustment rod.

[0018] The figures are not to scale. Instead, the thickness of the layers or regions may be enlarged in the drawings. In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts.

[0019] Descriptors “first,” “second,” “third,” etc. are used herein when identifying multiple elements or components that may be referred to separately. Unless otherwise specified or understood based on their context of use, such descriptors are not intended to impute any meaning of priority or ordering in time but merely as labels for referring to multiple elements or components separately for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for ease of referencing multiple elements or components.

DETAILED DESCRIPTION

[0020] Bicycles are known to have front forks that function as a suspension component. For example, a front fork typically includes a crown, a steerer tube extending upward from the crown, and two legs extending downward from the crown. Each leg has a first tube, in one example an upper cylindrical tube that is coupled to the crown and second tube, in one example a lower cylindrical tube that is to be connected to the front wheel. The upper and lower cylindrical tubes are arranged in a telescopic relationship that enables the front fork to compress and expand to absorb shocks and vibrations. Some front forks include spring and damper systems. For example, some front forks include a damper incorporated into one of the legs and a spring (e.g., an air spring, a coil spring) incorporated into the other leg. The spring enables the front fork to compress or contract when riding over a bump or obstacle, thereby reducing the transmission of shocks and vibrations to the rider, and then returns the fork to an expanded state after the compressive force is removed. The damper controls the speed at which the fork compresses and expands.

[0021] The upper and lower tubes of the front fork are moveable between a fully extended position and a fully compressed position, often referred to as top-out and bottom-out positions. These positions are typically defined by hard stops in which the front fork can no longer be expanded or compressed. During large compression events, such as when riding over large obstacles or landing after being airborne, the front fork can be fully compressed to its bottom-out position. This can feel harsh or jarring to the rider. Therefore, some front forks include a bumper, sometimes referred to as a jounce bumper or a bump stop. The jounce bumper is constructed of an elastomeric and/or compressible material (that is elastically deformable and/or compressible. For example, the jounce bumper may be constructed of a microcellular or foam material such as microcellular urethane. The jounce bumper is used to reduce or prevent the abrupt and harsh bottoming-out of the front fork when under full compression. Ideally the jounce bumper absorbs impact and dampens noise, vibration, and harshness by preventing the front fork from fully compacting during shock impacts caused by heavy loads (jumps), potholes, curbs, or objects (rocks). Jounce bumpers are typically integrated into the spring design for the end of travel compression, where the jounce bumper begins

to ramp up force. A jounce bumper affects the compression spring curve as well as the rebound curve. One advantage of using a jounce bumper instead of a spring (e.g., a coil spring) is that the jounce bumper does not rebound nearly as quickly (i.e., exhibits a hysteresis effect). Known jounce bumpers that are installed into forks are placed at a single location and cannot be affected or changed by the end user.

[0022] Disclosed herein are example front forks with adjustable jounce bumpers. In particular, the position of the jounce bumper can be adjusted within the front fork, which affects the position at which the jounce bumper is initialized, and the upper and lower tubes start to reduce speed before reaching the bottom-out position. The adjustability affects the compression amount and impacts the rate and force of the jounce bumper. This adjustability also enables a user to control and modify the end of the spring curve, which is beneficial for achieving the rider's preferred feedback during bottoming-out events.

[0023] In some examples disclosed herein, the jounce bumper is moveable vertically (axially) within one of the legs of the front fork. In some examples, the jounce bumper position can be adjusted by rotating an adjuster pin on the bottom end of the leg. The adjuster pin can be rotated by hand without a tool or with a tool (e.g., a hex tool or Allen wrench). The example adjustable jounce bumpers disclosed herein can be incorporated into the spring side, the damper side, or both the spring and damper sides.

[0024] Turning now to the figures, FIG. 1 illustrates one example of a human powered vehicle on which the example suspension components disclosed herein may be implemented. In this example, the vehicle is one possible type of bicycle **100**, such as a mountain bicycle. In the illustrated example, the bicycle **100** includes a frame **102** and a front wheel **104** and a rear wheel **106** rotatably coupled to the frame **102**. In the illustrated example, the front wheel **104** is coupled to the front end of the frame **102** via a front fork **108**. A front and/or forward riding direction or orientation of the bicycle **100** is indicated by the direction of the arrow A in FIG. 1. As such, a forward direction of movement for the bicycle **100** is indicated by the direction of arrow A. The terms “upper,” “lower,” “bottom,” “top,” “rear,” “front,” “fore,” “aft,” “vertical,” “horizontal,” “right,” “left,” “inboard,” “outboard” and variations or derivatives thereof, refer to the orientations of the exemplary bicycle **100**, shown in FIG. 1, from the perspective of a user who is seated on the bicycle **100** facing handlebars **114**.

[0025] In the illustrated example of FIG. 1, the bicycle **100** includes a seat **110** coupled to the frame **102** (e.g., near the rear end of the frame **102** relative to the forward direction A) via a seat post **112**. The bicycle **100** also includes the handlebars **114** coupled to the front fork **108** (e.g., near a forward end of the frame **102** relative to the forward direction A) for steering the bicycle **100**. The bicycle **100** is shown on a riding surface **116**. The riding surface **116** may be any riding surface such as the ground (e.g., a dirt path, a sidewalk, a street, etc.), a man-made structure above the ground (e.g., a wooden ramp), and/or any other surface.

[0026] In the illustrated example, the bicycle **100** has a drivetrain **118** that includes a crank assembly **120**. The crank assembly **120** is operatively coupled via a chain **122** to a sprocket assembly **124** mounted to a hub **126** of the rear wheel **106**. The crank assembly **120** includes at least one, and typically two, crank arms **128** and pedals **130**, along with at least one front sprocket, or chainring **132**. A rear gear change device **134**, such as a derailleur, is disposed at the rear wheel **106** to move the chain **122** through different sprockets of the sprocket assembly **124**. Additionally or alternatively, the bicycle **100** may include a front gear change device to move the chain **122** between a plurality of chainrings **132**.

[0027] The example bicycle **100** includes a suspension system having one or more suspension components. In this example, the front fork **108** is implemented as a front suspension component. The front fork **108** is or integrates a shock absorber that includes a spring and a damper, disclosed in further detail herein. Further, in the illustrated example, the bicycle **100** includes a rear suspension component **136**, which is a shock absorber, referred to herein as the rear shock absorber

136. The rear shock absorber **136** is coupled between two portions of the frame **102**, including a swing arm **138** coupled to the rear wheel **106**. The front fork **108** and the rear shock absorber **136** absorb shocks and vibrations while riding the bicycle **100** (e.g., when riding over rough terrain). In other examples, the front fork **108** and/or the rear shock absorber **136** may be integrated into the bicycle **100** in other configurations or arrangements. Further, in other examples, the suspension system may employ only one suspension component (e.g., only the front fork **108**) or more than two suspension components (e.g., an additional suspension component on the seat post **112**) in addition to or as an alternative to the front fork **108** and rear shock absorber **136**.

[0028] While the example bicycle **100** depicted in FIG. **1** is a type of mountain bicycle, the example suspension components (e.g., a front fork) with jounce bumpers disclosed herein can be implemented on other types of bicycles. For example, the disclosed suspension components with jounce bumpers may be used on road bicycles, as well as bicycles with mechanical (e.g., cable, hydraulic, pneumatic, etc.) and non-mechanical (e.g., wired, wireless) drive systems. The disclosed suspension components with jounce bumpers may also be implemented on other types of two-wheeled, three-wheeled, and four-wheeled human powered vehicles. Further, the example suspension components with jounce bumpers can be used on other types of vehicles, such as motorized vehicles (e.g., a motorcycle, a car, a truck, etc.).

[0029] FIG. **2** is a perspective view of an example front fork **200** (a suspension component) that can be implemented as the front fork **108** and used on the bicycle **100** of FIG. **1**. In the illustrated example of FIG. **2**, the front fork **200** includes a steerer tube **202**, a crown **204**, a first leg **206**, and a second leg **208**. Alternatively, the fork **200** may only include one leg. The steerer tube **202** is coupled to the crown **204** and extends outward (e.g., upward) from a top of the crown **204**. The steerer tube **202** is to be inserted through a head tube on the frame **102** (FIG. **1**) of the bicycle **100** and coupled (e.g., via a stem) to the handlebars **114** (FIG. **1**). The first and second legs **206**, **208** are coupled to the crown **204** and extend outward (e.g., downward) from a bottom of the crown **204**, opposite the steerer tube **202**. The first and second legs **206**, **208** are to be coupled to the front wheel **104** (FIG. **1**).

[0030] In the illustrated example, the first and second legs **206**, **208** include first and second upper tubes **210**, **212**, respectively, and first and second lower tubes **214**, **216**, respectively. The upper and lower tubes **210**, **212**, **214**, **216** are sometimes referred to as stanchions or leg portions. The first and second upper tubes **210**, **212** are coupled to and extend downward from the crown **204**. The front fork **200** includes an arch **218** (sometimes referred to as a fork brace or stabilizer) coupled between the lower tubes **214**, **216**. In some instances, the upper tubes **210**, **212** are referred to as an upper tube assembly, while the lower tubes **214**, **216** and the arch **218** are referred to as a lower tube assembly. The first and second lower tubes **214**, **216** include respective front wheel attachment portions **220**, **222**, such as holes (e.g., eyelets) or dropouts, for attaching the front wheel **104** (FIG. **1**) to the front fork **200**. In this example the upper and lower tubes **210**, **212**, **214**, **216** are cylindrical, but in other examples can have a different cross-sectional shape.

[0031] The first and second upper tubes **210**, **212** are slidably received within the respective first and second lower tubes **214**, **216**. Thus, the first and second upper tubes **210**, **212** form a telescopic arrangement with the respective first and second lower tubes **214**, **216**. During a compression stroke, the first and second upper tubes **210**, **212** move into or toward the respective first and second lower tubes **214**, **216**, and during a rebound stroke, the first and second upper tubes **210**, **212** move out of or away from the respective first and second lower tubes **214**, **216**.

[0032] FIG. **3** is a cross-sectional view of the example front fork **200** taken along line A-A of FIG. **2**. As shown in FIG. **3**, the first upper tube **210** has a first end **300**, referred to herein as a top end **300**, and a second end **302**, referred to herein as a bottom end **302**, opposite the top end **300**. The top end **300** is coupled to the crown **204**. In the illustrated example, a portion of the first upper tube **210** extends into an opening **304** in the crown **204**. In some examples, the first upper tube **210** is coupled to the crown **204** by a cap **305** that is threadably coupled to the top end **300** of the first

upper tube **210**. Additionally or alternatively, the first upper tube **210** can be coupled to the crown **204** via another mechanical and/or chemical fastening technique (e.g., threaded fasteners, welding, an adhesive, etc.). The first lower tube **214** has a third end **306**, referred to herein as a top end **306**, and a fourth end **308**, referred to herein as a bottom end **308**, opposite the top end **306**. The first upper tube **210** is inserted into the first lower tube **214**. In particular, the bottom end **302** of the first upper tube **210** is disposed within the first lower tube **214**. This type of configuration is sometimes referred to as a right side up fork. Alternatively, the first lower tube is inserted into the first upper tube. This type of configuration is sometimes referred to as an inverter fork. The top end **300** of the first upper tube **210** and the bottom end **308** of the first lower tube **214** form first and second distal ends of the suspension component. During compression, the top end **300** and the bottom end **308** are moved toward each other, and during extension or rebound, the top end **300** and the bottom end **308** are moved away from each other. Thus, the first upper and lower tubes **210**, **214** form a telescopic arrangement. The first upper and lower tubes **210**, **214** are moveable along a central axis **310** of the first leg **206**. The first upper and lower tubes **210**, **214** define an interior chamber or region **312**.

[0033] The second upper and lower tubes **212**, **216** are similarly arranged. In particular, the second upper tube **210** has a first end **314**, referred to herein as a top end **314**, and a second end **316**, referred to herein as a bottom end **316**, opposite the top end **314**. The top end **314** is coupled to the crown **204**. In the illustrated example, a portion of the second upper tube **212** extends into an opening **318** in the crown **204**. In some examples, the second upper tube **212** is coupled to the crown **204** by a cap **320** that is threadably coupled to the top end **314** of the second upper tube **212**. Additionally or alternatively, the second upper tube **212** can be coupled to the crown **204** via another mechanical and/or chemical fastening technique (e.g., threaded fasteners, welding, an adhesive, etc.). The second lower tube **216** has a third end **322**, referred to herein as a top end **322**, and a fourth end **324**, referred to herein as a bottom end **324**, opposite the top end **322**. The second upper tube **212** is inserted into the second lower tube **216**. In particular, the bottom end **316** of the second upper tube **212** is disposed within the second lower tube **216**. The second upper and lower tubes **212**, **216** form a telescopic arrangement and move along a central axis **326** of the second leg **208**. The second upper and lower tubes **212**, **216** define an interior chamber or region **328**. The first and second upper tubes **210**, **212** and the first and second lower tubes **214**, **216** are moveable between a fully extended position (also referred to as a top-out position) and a fully compressed position (also referred to as a bottom-out position).

[0034] In the illustrated example, the front fork **200** includes both a spring **330** and a damper **332**. In this example, the spring **330** is disposed in and/or otherwise integrated into the first leg **206**, and the damper **332** is disposed in and/or otherwise integrated into the second leg **208**. The spring **330** is configured to resist compression of the top ends **300**, **314** toward the bottom ends **308**, **324** and return the tubes **210**, **212**, **214**, **216** to the extended position after compression occurs. The damper **332** is configured to limit the speed at which the compression/extension occurs and/or otherwise absorb vibrations.

[0035] In this example, the spring **330** is implemented as an air spring. The spring **330** includes a pneumatic chamber **334**. In this example, the pneumatic chamber **334** is defined by an interior of the first upper tube **210**. However, in other examples, the pneumatic chamber **334** can be defined by a separate cylinder or body that is disposed in the first upper tube **210**, similar to the damper body **348** in the first upper tube **210**. The front fork **200** includes a seal head **336** (e.g., a piston, a plug, a disc, etc.) coupled to and disposed in the first upper tube **210** near the bottom end **302** that seals the bottom of the pneumatic chamber **334**. In this example, the steelhead **336** is spaced from the bottom end **302** of the first upper tube **210**. The cap **305** seals the top of the pneumatic chamber **334**. In some examples, the cap **305** includes a valve **338** (e.g., a Schrader valve) that can be used to fill the pneumatic chamber **334** with fluid (e.g., compressed air).

[0036] In the illustrated example, the spring **330** includes a first shaft **340** that is coupled to and

extends upward from the bottom end **308** of the first lower tube **214**. The first shaft **340** extends through the seal head **336** and into the pneumatic chamber **334**. The spring **330** includes a piston **342** that is coupled (e.g., threadably coupled) to an end of the first shaft **340** and disposed in the pneumatic chamber **334** in the first upper tube **210**. The piston **342** is slidable within the first upper tube **210**. In some examples, a seal is disposed around the piston **342**, which creates a seal between the piston **342** and the inner surface of the first upper tube **210**.

[0037] The piston **342** divides the pneumatic chamber **334** into a first chamber **344**, referred to herein as a positive air chamber **344**, and a second chamber **346**, referred to herein as a negative air chamber **346**. In some examples, the positive air chamber **344** is filled with a mass of a pneumatic fluid (e.g., a gas, such as air) having a higher pressure than ambient pressure. Therefore, in this example, the positive air chamber **344** forms a pressurized chamber, sometimes referred to as a highly pressurized zone or positive spring chamber, above the piston **342**. The negative air chamber **346** is also filled with pneumatic fluid and forms a negative spring chamber below the piston **342**. When the front fork **200** compresses and the ends of the first upper and lower tubes **210**, **214** move toward each other, such as when riding over a bump, the first shaft **340** moves the piston **342** toward the top end **300** of the first upper tube **210**. As a result, the volume of the positive air chamber **344** decreases and, thus, the pressure of the air within the positive air chamber **344** increases. Conversely, the volume of the negative air chamber **346** increases and therefore the pressure of the air in the negative air chamber **346** decreases. After the compressive force is removed, the increased pressure in the positive air chamber **344** and the decreased pressure in the negative air chamber **346** acts to move the piston **342** away from the top end **300**, which pushes the ends of the first upper and lower tubes **210**, **214** away from each other, thereby acting as a spring to return the front fork **200** to its original or riding set up. The second upper and lower tubes **212**, **216** similarly follow this motion.

[0038] In other examples, the spring **330** can be implemented by a physical spring, such as a coil spring. For example, a coil spring can be disposed in the first upper tube **210** between the first shaft **340** and the top end **300** of the first upper tube **210**. When the front fork **200** is compressed, the first shaft **340** is moved upward and compresses the coil spring. After the compression, the coil spring acts to expand the front fork **200** back to its original or riding set up.

[0039] In the illustrated example, the damper **332** includes a damper body **348** that defines a chamber **350** (e.g., a hydraulic chamber). The damper body **348** is disposed in and coupled to the second upper tube **212**. For example, the damper body **348** is coupled to and extends downward from the second cap **320**. As such, the damper body **348** is coupled to and disposed in a fixed position in the second upper tube **212**. In other examples, the chamber **350** can be formed by the interior of the second upper tube **212**. The bottom of the chamber **350** is sealed by a seal head **352**. The chamber **350** is filled with fluid. The fluid may be, for example, oil, such as a mineral oil based damping fluid. In other examples, other types of damping fluids may be used (e.g., silicone or glycol type fluids). The damper **332** includes a second shaft **354** (which may be referred to as a damper or piston shaft, rod, or stem). The second shaft **354** is coupled to and extends upward from the bottom end **324** of the second lower tube **216**. The second shaft **354** extends upward and through the seal head **352** and into the chamber **350**. The damper **332** includes a damper member **356** (which may also be referred to as a piston or mid-valve) disposed in the chamber **350** of the damper body **348**. The damper member **356** is coupled to the second shaft **354** and is slidable in the damper body **348**. The damper member **356** divides the chamber **350** into two chambers (above and below the damper member **356**). When the front fork **200** compresses and the ends of the second upper and lower tubes **212**, **216** move toward each other, such as when riding over a bump, the second shaft **354** moves the damper member **356** upward in the chamber **350** toward the top end **314** of the second upper tube **212**. During rebound, the damper member **356** moves downward in the chamber **350** away from the top end **314** of the second upper tube **212**. The damper member **356** includes one or more channels that enable fluid to flow across the damper member **356**, at a

restricted rate, between the first and second chambers, thereby damping or slowing the compression/extension movement of the front fork **200**.

[0040] In some examples, the rebound and compression rates of the damper **332** can be independently controlled. For example, as shown in FIG. **3**, the damper **332** includes an accumulation chamber **358**. In some examples, when the front fork **200** is compressed, such as during a high compression event, the damper member **356** moves upward in the chamber **350** and forces the fluid through a top of the damper body **348** and into the accumulation chamber **358**. The damper **332** includes a spring **360** that biases a piston or diaphragm downward to apply pressure to the fluid in the accumulation chamber **358**. In some examples, the resistance of the spring **360** can be adjusted via a compression adjust rod. In some examples, the damper **332** can include one or more adjustment knobs or dials (e.g., on the second cap **320**) to adjust one or more parameters of the piston/diaphragm and/or spring **360** to control the rebound and/or compression damping rates.

[0041] The spring **330** and the damper **332** include multiple seals. These seals have a static friction that must be overcome to compress or expand the front fork **200**. While relatively small, this static friction may cause a delay in the compression or rebound movement, which may cause an undesirable stick slip feeling that can be felt by the rider. Additionally, high frequency vibrations (e.g., above 5 Hz) having a low amplitude may be not absorbed by the spring **330** and the damper **332**. Therefore, in some examples, the front fork **200** includes a first example isolator **362** to address the above-noted drawbacks. In this example, the first isolator **362** is associated with the spring **330** in the first leg **206**. The first isolator **362** is disposed between and couples the first shaft **340** to the bottom end **308** of first lower tube **214**. The first isolator **362** enables relative movement between the first lower tube **214** (which is attached to the front wheel **104** (FIG. **1**) and considered the unsprung side of the suspension component) and the first shaft **340**, which is coupled to the piston **342**. In particular, as disclosed in further detail herein, the first isolator **362** includes one or more cushioning members, such as elastomeric members (e.g., rubber pads). The elastomeric member(s) of the first isolator **362** enable(s) relative movement between the first lower tube **214** and the first shaft **340** and, thus, between the first upper and lower tubes **210**, **214**. As such, the first isolator **362** enables the first lower tube **214** (the unsprung mass) to move upward relative to the first upper tube **210** before the breakaway force for the spring **330** and the damper **332** is reached, thereby enabling the front fork **200** absorb the vibrations more quickly during compression. The first isolator **362** also absorbs high frequency, low amplitude vibrations that would otherwise be transmitted through the first upper and lower tubes **210**, **214** to the handlebars **114** (FIG. **1**). In the illustrated example, the front fork **200** also includes a second isolator **364** associated with the damper **332** in the second leg **208**. The second isolator **364** is disposed between and couples the second shaft **354** to the bottom end **324** of the second lower tube **216**. The second isolator **380** is substantially the same as the first isolator **362** and similarly allows relative movement between the second lower tube **216** and the second shaft **354** and, thus, between the second upper and lower tubes **212**, **216**. In the illustrated example, the front fork **200** includes two isolators, one in each of the legs **206**, **208**. However, in other examples, the front fork **200** may only include one isolator (e.g., only the first isolator **362**).

[0042] In the illustrated example, the front fork **200** includes a bumper, in this example a jounce bumper **366**. The jounce bumper **366** is disposed in the first lower tube **214** of the first leg and, in particular, near the bottom end **308** of the first lower tube **214**. The jounce bumper **366** may be constructed of an elastomeric material (e.g., rubber) and is used to cushion the impact during a bottoming-out event. In particular, when the front fork **200** is compressed during a high-speed or high-force compression event, the seal head **336** and/or the bottom end **302** of the first upper tube **210** engages the jounce bumper **366**. The jounce bumper **366** deforms and/or compresses before stopping the first upper tube **210** from moving further downward. As such, the jounce bumper **366** helps to reduce or soften the hard stopping point when the front fork **200** is fully compressed. The jounce bumper **366** also affects the end of the spring curve for the spring **330**. As disclosed in

further detail herein, the position of the jounce bumper 366 is adjustable in the first lower tube 214. For example, the jounce bumper 366 can be moved axially (e.g., up or down in FIG. 3) relative to the first lower tube 214 by actuating a bumper adjuster 365. This enables a user to adjust and/or otherwise control the end spring rate during a bottoming-out event of the front fork 200.

[0043] FIG. 4A is an exploded view of the assembly of parts in the first leg 206 including the first shaft 340, the seal head 336, the first isolator 362, the jounce bumper 366 and the bumper adjuster 365. FIG. 4B is a cross-sectional view of the same parts in the exploded view of FIG. 4A. As shown in FIGS. 4A and 4B, the assembly includes O-rings 402, 403, 404, the seal head 336, a bushing 405, an O-ring 407, a seal head spacer 409, the jounce bumper 366, a bushing 406, a bumper carrier 408, an upper isolator body 410, a first cushioning member 412, a coupler 414, a second cushioning member 416, an O-ring 418, a lower isolator body 420, a set screw 422, an O-ring 424, a spring 426, a ball 428, an O-ring 430, a bumper actuator 432, an O-ring 434, a bottom nut 436, and a bottom-out pad 438. These parts are disclosed in further detail herein.

[0044] FIG. 5A is an enlarged view of the callout 368 of FIG. 3 showing the first isolator 362 and the jounce bumper 366 in the first lower tube 214. The FIG. 5A shows the jounce bumper 366 in a first position, in this example, a lower-most position. FIG. 5B, which is described in further detail below, shows the jounce bumper 366 moved to a second position, in this embodiment, an upper-most position. The lower-most position and upper-most position may also be referred to as minimum and maximum positions, respectively. The jounce bumper 366 is moveable to a plurality of positions between the lower-most position (FIG. 5A) and the upper-most position (FIG. 5B) by the bumper adjuster 365. The bumper adjuster 365 generally comprises the bumper carrier 408 and the bumper actuator 432.

[0045] As shown in FIG. 5A, the bottom-out pad 438 is disposed on a shoulder 500 formed in the first lower tube 214. The bottom-out pad 438 forms a hard stop when the first upper and lower tubes 210, 214 are fully compressed, as shown in further detail herein. In some examples, the bottom-out pad 438 is constructed of rubber. In some examples, the bottom-out pad 438 includes rubber molded onto a plastic bottom.

[0046] In the illustrated example of FIG. 5A, the first lower tube 214 has an opening 502 extending between a first side 504 (an internal side) and a second side 506 (an external side) of the bottom end 308. The lower isolator body 420 is disposed in the interior region 312 of the first lower tube 214. The O-ring 424 is disposed in a gland in an upper portion 514 of the lower isolator body 420 and forms a seal between the lower isolator body 420 and the first side 504 of the bottom end 308 to prevent leakage of fluid (e.g., air, oil, etc.). The lower isolator body 420 has a lower portion 508 that extends through the opening 502. The lower portion 508 has external threads 507. The bottom nut 436 has internal threads 435 and is threadably coupled to the lower portion 508 on the lower isolator body 420, which thereby couples the lower isolator body 420 to the bottom end 308 of the first lower tube 214. The bottom nut 436 can be torqued to rigidly secure the lower isolator body 420 to the first lower tube 214. The O-ring 434 is disposed in a gland in the bottom nut 436 and forms a seal (e.g., to block dust) between the bottom nut 436 and the second side 506 of the bottom end 308. In this example, the bumper actuator 432 is implemented as a pin, referred to herein as an adjuster pin 432. The adjuster pin 432 extends through a central passage 510 in the lower isolator body 420 and, thus, through the opening 502 in the bottom end 308 of the first lower tube 214. The adjuster pin 432 is accessible by a user from the bottom of the first lower tube 214. The adjuster pin 432 is rotatable in the lower isolator body 420, which causes rotation of the bumper carrier 408, as disclosed in further detail herein. As shown in FIG. 5A, the set screw 422 is screwed into the lower isolator body 420 and extends into a groove 512 on the side of the adjuster pin 432. This allows the adjuster pin 432 to rotate in the lower isolator body 420, but prevents the adjuster pin 432 from moving axially (e.g., up or down) relative to the lower isolator body 420. In the illustrated example, the O-ring 418 is disposed in a gland in the adjuster pin 432 and forms a seal between the adjuster pin 432 and the inner surface of the lower isolator body 420 to prevent leakage of fluid (e.g., air,

oil, etc.).

[0047] As shown in FIG. 5A, the upper isolator body **410** is threadably coupled to the lower isolator body **420**. The upper portion **514** of the lower isolator body **420** has internal threads **421**, and a lower portion **413** of the upper isolator body **410** has external threads **411** and is threadably coupled to the upper portion **514** of the lower isolator body **420**. In other examples, the upper and lower isolator bodies **410**, **420** can be coupled via other mechanical and/or chemical fastening techniques (e.g., welding, an adhesive, a threaded fastener such as a screw or bolt, etc.). The upper and lower isolator bodies **410**, **420** are coupled together and form a body **515** (e.g., a housing) that is rigidly coupled to the bottom end **308** of the first lower tube **214**. The upper and lower isolator bodies **410**, **420** define a cavity **516**. As shown in FIG. 5A, the first and second cushioning members **412**, **416** are disposed in the cavity **516** and clamped (e.g., axially constrained) between the upper and lower isolator bodies **410**, **420**.

[0048] In this example, the coupler **414** is implemented as a bolt **414**. Looking to FIG. 5A, the bolt **414** has a plate portion **518** (e.g., a flange, a disk) and a post portion **520** with a threaded section **522**. The plate portion **518** is disposed (e.g., clamped) between the first and second cushioning members **412**, **416** in the cavity **516**, and the post portion **520** extends outward through an opening **524** in the upper isolator body **410**. The threaded section **522** is threaded into the first shaft **340** with the O-ring **402** forming a seal between the bolt **414** and the first shaft **340**. As such, the bolt **414** and the first shaft **340** are rigidly coupled and move together. In the illustrated example, the bolt **414** has a bore **526**. As shown in FIG. 5A, the adjuster pin **432** extends into the bore **526** of the bolt **414**. The bolt **414** is movable (e.g., slidable) axially along the adjuster pin **432**. As such, the bolt **414** and the first shaft **340** are moveable axially relative to the first lower tube **214**. The adjuster pin **432** and bolt **414** are rotationally keyed. In particular, the adjuster pin **432** and the bore **526** have a matching cross-sectional shape, such as a hexagon. This allows the adjuster pin **432** and the bolt **414** to move (e.g., slide) axially relative to each other, but rotationally fixes the adjuster pin **432** and the bolt **414** such that rotation of the adjuster pin **432** causes rotation of the bolt **414**.

[0049] In this example, the first and second cushioning members **412**, **416** are implemented as elastomeric members. The first and second cushioning members **412**, **416** can be constructed of any elastomeric material. In some examples, the first and second cushioning members **412**, **416** are constructed of nitrile rubber (e.g., 40 Shore A nitrile rubber). In other examples, the first and second cushioning members **412**, **416** can be constructed of other types of rubber (e.g., butyl rubber, ethylene propylene diene monomer (EPDM) rubber, etc.), silicone, polyurethane, or a viscoelastic material. In this example, the first and second cushioning members **412**, **416** are ring-shaped. However, in other examples, the first and second cushioning members **412**, **416** can be shaped differently.

[0050] The first and second cushioning members **412**, **416** are engaged with opposite sides of the plate portion **518**. Therefore, the first cushioning member **412** biases the plate portion **518** (and, thus, the bolt **414** and the first shaft **340**) downward, and the second cushioning member **416** biases the plate portion **518** in the opposite direction. In some examples, the first and second cushioning members **412**, **416** are preloaded (i.e., in a slightly deformed and/or compressed state).

[0051] The first and second cushioning members **412**, **416** elastically deform and/or compress, and expand, in response to compression and rebound forces. For example, when a compressive force is first applied to the front fork **200** (e.g., when riding over a bump), the upper and lower isolator bodies **410**, **420** are forced upward and/or the bolt **414** is forced downward. Before the breakaway force is reached, the second cushioning member **416** is deformed and/or compressed between the lower isolator body **420** and the plate portion **518** of the bolt **414**, which enables the first lower tube **214** to move upward relative to the first shaft **340** and, thus, upward relative to the first upper tube **210** (FIG. 2). Further, because the plate portion **518** is moved away from the first cushioning member **412**, the first cushioning member **412** expands. After the compressive force is removed, the second cushioning member **416** biases the lower isolator body **420** and the plate portion **518**

away from each other, which moves the first lower tube **214** downward relative to the first shaft **340** and, thus, downward relative to the first upper tube **210**. Similarly, when a rebound (expanding) force is applied to the front fork **200** (e.g., from the spring **330** (FIG. 3)), the first and second cushioning members **412**, **416** enable relative movement of the first upper and lower tubes **210**, **214** in the opposite direction. In this manner, the first isolator **362** enables relative movement between the first upper and lower tubes **210**, **214** before the breakaway forces of the spring **330** and the damper **332** (FIG. 3) are reached. Therefore, unlike known front forks, the example front fork **200** does not require a certain force to overcome some friction or breakaway force to initiate movement. Instead, any net compressive or expansive force can result in relative movement of the first upper and lower tubes **210**, **214**. This results in less vibrations or shocks transmitted through the front fork **200** to the handlebars **114** (FIG. 1). The first and second cushioning members **412**, **416** also absorb high frequency, low amplitude vibrations that may otherwise not be absorbed by the front fork **200**. In this example the first isolator **362** include two cushioning members.

However, in other examples, only one cushioning member may be implemented. For example, in some instances, only the first cushioning member **412** may be included. Further, while in this example the cushioning members are implemented as elastomeric members, in other examples, the cushioning members can be implemented as springs (e.g., metallic coil springs, leaf springs, etc.) or other types of cushioning members that produce biased movement between two components.

[0052] As shown in FIG. 5A and 5B, the bumper carrier **408** is disposed in the first lower tube **214**. The bumper carrier **408** can also be referred to as a bumper cup. The bumper carrier **408** is cylindrical. An interior of the bumper carrier **408** includes a flange **531** dividing the bumper carrier into a lower portion **528** and an upper portion **530**. The bumper carrier **408** is threadably coupled to the body **515** of the first isolator **362**. In particular, the lower portion **528** of the bumper carrier **408** includes internal threads **527** that are engaged with the external threads **529** on the upper isolator body **410**. The jounce bumper **366** is coupled to the bumper carrier **408**. For example, as shown in FIG. 5A, the upper portion **530** and the flange **531** form a cup shape defining a cavity **532** for receiving the jounce bumper **366**. The jounce bumper **366** is disposed in the cavity **532** and contacts the flange **531**. In some examples, the jounce bumper **366** is coupled to the bumper carrier **408** via friction fit with the upper portion **530**. Additionally or alternatively, the jounce bumper **366** can be coupled to the bumper carrier **408** via other mechanical and/or chemical techniques (e.g., an adhesive, a threaded fastener, etc.). As disclosed in further detail herein, the bumper carrier **408** is axially moveable to adjust the position the jounce bumper **366** in the first lower tube **214**.

[0053] When the front fork **200** is compressed, the first upper tube **210** (FIG. 3) is moved into the first lower tube **214**. If the compressive force is large enough, the first upper tube **210** is moved relatively far into the first lower tube **214** such that the seal head **336** (FIG. 3) contacts the jounce bumper **366**. The seal head **336** contacts the jounce bumper **366** first, which deforms and/or compresses the jounce bumper **366** until the first upper tube **210** contacts the bottom-out pad **438**. Therefore, in some examples, the engagement with the bottom-out pad **438** defines the fully compressed or bottom-out position of the front fork **200**. However, in other examples, the bottom-out pad **438** is not contacted. Instead, the jounce bumper **366** is deformed and/or compressed until its thickness cannot be forcibly reduced anymore, which prevents the first upper tube **210** from moving further into the first lower tube **214** and, thus, defines the fully compressed or bottom-out position of the front fork **200**. In either instance, this deformation and/or compression of the jounce bumper **366** provides a more gradual slowing of the tubes **210**, **214** before the hard stop at the bottom-out position. As such, the jounce bumper **366** prevents or reduces the abrupt or harsh bottoming-out of the front fork **200**.

[0054] For example, referring briefly to FIG. 6A, FIG. 6A shows the jounce bumper **366** in the lower-most position (FIG. 5A) during a bottoming-out event when the seal head **336** makes initial contact with the jounce bumper **366**. In the upper-most position, the top of the jounce bumper **366** is a first distance **D1** from the bottom end **308** of the first lower tube **214**. As the first upper tube

210 continues to move downward, the jounce bumper **366** is deformed and/or compressed between the seal head **336** and the bumper carrier **408**. In some examples, the jounce bumper **366** is constructed of an elastomeric and/or compressible material such as microcellular or foam material such as microcellular urethane (MDU), rubber (e.g., nitrile rubber, butyl rubber, ethylene propylene diene monomer (EPDM) rubber, etc.), silicone, and/or polyurethane. The first upper tube **210** continues to move downward until the deformation/compression of the jounce bumper **366** stops the movement and/or the bottom end **302** of the first upper tube **210** contacts the bottom-out pad **438**, as shown in FIG. 6B. This contact with the bottom-out pad **438** provides a hard stop that prevents the first upper tube **210** from moving further into the first lower tube **214**. As shown in FIG. 6B, the jounce bumper **366** has been deformed and/or compressed (in the axial direction) a first amount C1 from the expanded state in FIG. 6A. The deformation and/or compression of the jounce bumper **366** over the first amount C1 provides a gradual slowing of the first upper tube **210** before reaching the hard stop.

[0055] As shown in FIG. 6B, the O-ring **404** is carried by the seal head **336** and seals against the first shaft **340** to prevent fluid leakage between the seal head **336** and the first shaft **340**. The bushing **405** enables the seal head **336** to slide relative to the first shaft **340**. Further, the O-ring **403** is carried by the seal head **336** and seals against the inner surface of the first upper tube **210** to prevent fluid leakage between the seal head **336** and the first upper tube **210**. In the illustrated example, the seal head spacer **409** is disposed in the first upper tube **210**. The seal head spacer **409** is secured in the first upper tube **210** by a retainer **602** (e.g., a circlip). The seal head spacer **409** maintains the seal head **336** at a certain position in the first upper tube **210** and prevents the seal head **336** from being ejected from the bottom end **302** of the first upper tube **210**. The O-ring **407** is clamped between the seal head **336** and the seal head spacer **409**.

[0056] Referring back to FIG. 5A, the position of the jounce bumper **366** can be changed to adjust the position at which the seal head **336** (FIG. 3) contacts the jounce bumper **366** during a bottoming-out event. As shown in FIG. 5A, the post portion **520** of the of the bolt **414** extends through an opening **534** in the bumper carrier **408**. The bushing **406** is disposed in the opening **534** and coupled to the flange **531** of the bumper carrier **408** (e.g., via friction fit). The post portion **520** of the bolt **414** extends through the bushing **406**. The bushing **406** is rotationally keyed to the post portion **520** of the bolt **414**. As such, the bolt **414** and the bumper carrier **408** are rotationally key (e.g., rotationally fixed) while still allowing the bolt **414** and the bumper carrier **408** to move (e.g., slide) axially relative to each other. The bushing **406** forms a low friction surface for the bolt **414** to slide relative to the bumper carrier **408**. This reduces wear on the bolt **414** and the bumper carrier **408**. In some examples, the bushing **406** is constructed of polytetrafluorethylene (e.g., Teflon®). In other examples, the bushing **406** can be constructed of another material, such as polyoxymethylene or acetal (e.g., Delrin®).

[0057] To adjust the position of the jounce bumper **366**, a user can rotate the adjuster pin **432**. In some examples, the adjuster pin **432** can be rotated by inserting a tool (e.g., an Allen wrench) into a hex bore **536** and rotating the adjuster pin **432**. Additionally or alternatively, the adjuster pin **432** can be rotated by hand without a tool. In some examples, the adjuster pin **432** can have an enlarged knob shape that can be grasped by a user. As disclosed above, the adjuster pin **432** and the bolt **414** are keyed together, such that rotation of the adjuster pin **432** causes rotation of the bolt **414**. Further, the bolt **414** and the bumper carrier **408** are keyed together (e.g., via the bushing **406**). As such, rotation of the adjuster pin **432** causes rotation the bumper carrier **408**. The bumper carrier **408** is threadably engaged with the upper isolator body **410**, which is rigidly coupled to the first lower tube **214**. As such, rotation of the bumper carrier **408** causes the bumper carrier **408** and the jounce bumper **366** to move axially (e.g., up or down) relative to the upper isolator body **410** and, thus, relative to the first lower tube **214**. For example, FIG. 5B shows the bumper carrier **408** after being rotated and translated upward to an upper-most position. As shown, the bumper carrier **408** has been moved upward relative to the upper isolator body **410**. Therefore, the jounce bumper **366**

is spaced further from the bottom end **308** of the first lower tube **214**. As such, during a bottoming-out event, the seal head **336** (FIG. 3) engages the jounce bumper **366** at an earlier position, which affects the end of the spring curve. The axial movement of the bumper carrier **408** and the jounce bumper **366** is independent of the bolt **414** and the first shaft **340**. In other words, moving the bumper carrier **408** and the jounce bumper **366** up or down does not cause the bolt **414** or the first shaft **340** to move up or down. Instead, the bumper carrier **408** and the jounce bumper **366** slide linearly up or down the first shaft **340**. However, because the bolt **414** is rigidly coupled to the first shaft **340**, rotation of the bolt **414** also causes rotation of the first shaft **340** (and may also cause rotation of the piston **342** (FIG. 3). This rotation, however, does not affect the performance of the spring **330**.

[0058] Referring briefly to FIG. 7A, FIG. 7A shows the jounce bumper **366** in the upper-most position (FIG. 5B) during a bottoming-out event when the seal head **336** making initial contact with the jounce bumper **366**. In the upper-most position, the top of the jounce bumper **366** is a second distance **D2** from the bottom end **308** of the first lower tube **214**. The second distance **D2** is greater than the first distance **D1** of FIG. 6A. As such, the seal head **336** contacts the jounce bumper **366** at an earlier position in the downstroke. As the first upper tube **210** continues to move downward, the jounce bumper **366** is deformed and/or compressed between the seal head **336** and the bumper carrier **408**. The jounce bumper **366** is deformed and/or compressed until the bottom end **302** of the first upper tube **210** has contacted the bottom-out pad **438**, as shown in FIG. 7B. The first upper tube **210** continues to move downward until the deformation and/or compression of the jounce bumper **366** stops the movement and/or the bottom end **302** of the first upper tube **210** has contacted the bottom-out pad **438**, as shown in FIG. 7B. This contact with the bottom-out pad **438** provides a hard stop that prevents the first upper tube **210** from moving further into the first lower tube **214**. As shown in FIG. 7B, the jounce bumper **366** has been deformed and/or compressed a second amount **C2** from the expanded state in FIG. 7A. This second amount **C2** is greater than the first amount **C1** in FIG. 6B. Therefore, the jounce deformation/compression event occurs over a longer stroke when the jounce bumper **366** is in the upper-most position. This provides a more gradual damping before the fully compressed or bottom-out position. A user can adjust the position of the jounce bumper **366** to any position between the lower-most position (FIG. 5A) and the upper-most position (FIG. 5B) based on their desired preference.

[0059] As shown in FIG. 7B, the seal head **336** has a lower portion **700**. The lower portion **700** of the seal head **336** and the upper portion **530** of the bumper carrier **408** are used to radially constrain the jounce bumper **366**. In particular, when the jounce bumper **366** is deformed and/or compressed, the jounce bumper **366** is squeezed axially, which forces the jounce bumper **366** to expand radially. However, the jounce bumper **366** is radially constrained within the lower portion **700** of the seal head **336** and the upper portion **530** of the bumper carrier **408**. When the jounce bumper **366** has been maxed out or completely fills the space between the lower and upper portions **700**, **530**, the jounce bumper **366** acts as a hard material that stops further movement. As such, the inner diameter of the portions **700**, **530** and/or the length of the portions **700**, **530** controls the amount the jounce bumper **366** can be deformed and/or compressed and, thus, controls the spring rate of the jounce bumper **366**. The diameter of the portions **700**, **530** and/or the length of the portions **700**, **530** can be increased or decreased depending on the desired compression rate. In other examples, the seal head **336** and/or the bumper carrier **408** may not have these portions, which may result in a less stiff jounce bumper action. As shown in FIGS. 5A, 6A, and 7A, the jounce bumper **366** has a tapered profile (e.g., the top of the jounce bumper **366** is thinner) with one or more grooves in the outer radial surface of the jounce bumper **366**. This enables the jounce bumper **366** to fold at the thinner section(s). However, in other examples, the jounce bumper **366** can have other geometries that do not fold.

[0060] The jounce bumper **366** can be moved to any position between the lower-most position (FIG. 5A) and the upper-most position (FIG. 5B). In some examples, the front fork **200** includes

one or more features to limit the bumper carrier **408** from moving beyond the lower-most position (FIG. 5A) and the upper-most position (FIG. 5B). For example, as shown in FIG. 5A, the bumper carrier **408** is engaged with an upward extending portion **538** of the upper isolator body **410**. This prevents the bumper carrier **408** from being moved further downward and thereby defines the lower-most position of the jounce bumper **366**. As shown in FIG. 5B, the bushing **406** is engaged with a bottom end **540** of the first shaft **340**. This prevents the bumper carrier **408** from being moved further upward and thereby defines the upper-most position of the jounce bumper **366**. In other examples, other structures or features can be used to define these upper and lower limits of the bumper carrier **408**.

[0061] In some examples, the front fork **200** includes a position mechanism that defines one or more discrete positions between the lower-most position and the upper-most position. For example, as shown in FIG. 5A, the adjuster pin **432** has a flange **542** with multiple detents **544** (e.g., grooves, recesses) spaced apart and arranged circumferentially. Only one of the detents **544** is visible in FIG. 5A. As shown in FIG. 5A, the ball **428** and spring **426** are disposed in a bore **546** in the lower isolator body **420** and biased into one of the detents **544** on the flange **542** of the adjuster pin **432**. The ball **428** holds the adjuster pin **432** in its current rotational position until a sufficient torque is applied (e.g., by a user with a tool), at which point the ball **428** is pushed upward and the adjuster pin **432** can rotate freely. When another detent **544** is aligned with the ball **428**, the ball **428** slides into the detent **544** to hold the adjuster pin **432** in its current rotational position. Therefore, the adjuster pin **432** is rotatable to a plurality of discrete positions via a ball and detent mechanism, which defines one or more discrete positions of the jounce bumper **366**. The flange **542** can have any number of detents to define any number of discrete positions. In the illustrated example, the O-ring **430** forms a seal (e.g., to block dust) between the flange **542** and the lower isolator body **420**.

[0062] In this example, the bumper carrier **408** uses the upper isolator body **410** as a threaded structure to facilitate the up-down movement. Therefore, this design leverages the structure of the first isolator **362**. However, in other examples, the front fork **200** may not include the first isolator **362**. Instead, the first shaft **340** can be fixedly coupled to the bottom end **308** of the first lower tube **214** (e.g., via a threaded fastener). In such an example, the bumper carrier **408** may be threadably coupled to another threaded structure (e.g., a threaded sleeve or bolt) in the first lower tube **214**.

Therefore, the first isolator **362** is not a required element for the adjusting the jounce bumper **366**.

[0063] While in some examples the front fork **200** is configured such that the jounce bumper **366** is engaged by the seal head **336**, in other examples, the front fork **200** can be configured such that the bottom end **308** of the first upper tube **210** or another structure coupled to the first upper tube **210** contacts the jounce bumper **366**. Therefore, the jounce bumper **366** is contacted by the first upper tube **210** or a structure (e.g., the seal head **336**) coupled to the first upper tube **210** during a bottoming-out event before the first upper and lower tubes **210**, **214** reach the fully compressed positioned.

[0064] While the example adjustable jounce bumper **366** is described in connection with the first leg **206** having the spring **330**, a similar adjustable jounce can be implemented in the second leg **208** that has the damper **332**. Therefore, the example jounce bumper **366** is not limited to just the spring side. In some examples, the damper **332** may include an adjustment knob on the bottom end of the second leg **208** for adjusting a parameter of the damper **332**, such as the compression or rebounding damping rate. In such an instance, the adjustment knob is typically centrally located.

[0065] FIG. 8 is a cross-sectional view of an example jounce bumper **800** and bumper carrier **802** implemented in the second lower tube **216** of the second leg **208** (which incorporates the damper **332** (FIG. 3)). The second isolator **364**, the jounce bumper **800**, and the bumper carrier **802** operate in substantially the same manner as the first isolator **362**, the jounce bumper **366**, and the bumper carrier **408** described above and, thus, a description of these components is not repeated. As shown in FIG. 8, the front fork **200** includes a bolt **804** and an adjuster pin **806**, which are similar to the bolt **414** and the adjuster pin **432** disclosed above. However, in this example, the damper **332** (FIG.

3) includes a damper adjustment rod **808** that extends through central channels in the bolt **804** and the adjuster pin **806** and into the second shaft **354**. A bottom end **810** of the damper adjustment rod **808** is disposed below the adjuster pin **806**. The damper adjustment rod **808** can be rotated to adjust a fluid flow rate across the damper member **356** (FIG. 3), which affects the damping rate of the damper **332**. As shown in FIG. 8, a damper adjustment knob **812** is coupled to the damper adjustment rod **808**. The damper adjustment knob **812** is rotatably coupled to the adjuster pin **806**. As such, the damper adjustment knob **812** can rotate independent of the adjuster pin **806**. A user can rotate the damper adjustment knob **812** to rotate the damper adjustment rod **808**. In some examples, the damper adjustment knob **812** is removable to enable a user to access the adjuster pin **806** to adjust the position of the jounce bumper **800**. In the illustrated example, the front fork **200** includes a ball and detent mechanism to enable the damper adjustment knob to move to discrete positions. In the illustrated example, a flange **814** of the adjuster pin **806** has a plurality of detents **816** (one of which is visible in FIG. 8). The damper adjustment knob **812** has a bore **818** with a spring **820** and ball **822**. The ball **822** is biased into the detents **816** on the adjuster pin **806**. This allows the damper adjustment knob **812** (and, thus, the damper adjustment rod **808**) to be rotated to a plurality of discrete rotational positions.

[0066] While in some examples disclosed above the jounce adjustment is controlled externally, such as by rotating the adjuster pin **432**, **806**, in other examples, the position of the jounce bumper can involve an internal adjustment by the user. For example, a user may partially disassemble the front fork **200** and change the position of the jounce bumper **366**, **800** by manually rotating the bumper carrier **408**, **802** to move the position of the jounce bumper **366**, **800** up or down. In such an example, the front fork **200** may not include the adjuster pin **432**, **806**. In another example, the bolt **414**, **804** may have e-clip grooves, and a user can adjust the bumper carrier **408**, **802** up or down on the bolt **414**, **804** using a removable e-clip.

[0067] While the example adjustable jounce bumpers are described in connection with a front suspension fork, the example adjustable jounce bumpers disclosed herein can be similarly implemented in connection with other types of suspension components. For example, any of the example adjustable jounce bumpers disclosed herein can be similarly implemented in the rear shock absorber **136** (FIG. 1).

[0068] From the foregoing, it will be appreciated that example apparatus have been disclosed that enable a user to have adjustment of the bumper engagement and therefore the end stroke rate of the spring during a bottoming-out event.

[0069] Examples and combinations of examples disclosed herein include the following:

[0070] Example 1 is a front fork for a bicycle, the front fork comprising: a leg including a first tube and a second tube, the first tube having a first end and a second end, the second tube having a third end and a fourth end, the first and second tubes configured in a telescopic arrangement, the first and second tubes moveable between a fully extended position and a fully compressed position; and a bumper disposed in the second tube near the fourth end, the bumper to be contacted during a bottoming-out event before the first and second tubes reach the fully compressed positioned, wherein a position of the bumper is adjustable in the second tube.

[0071] Example 2 includes the front fork any preceding example, wherein the second end of the first tube is disposed within the second tube.

[0072] Example 3 includes the front fork any preceding example, wherein the second tube includes a wheel attachment portion extending from the fourth end, the wheel attachment portion to be coupled to a hub on a wheel of the bicycle.

[0073] Example 4 includes the front fork any preceding example, further including an adjuster comprising a bumper carrier in the second tube, the bumper coupled to the bumper carrier, wherein the bumper carrier is axially moveable to adjust the position the bumper in the second tube.

[0074] Example 5 includes the front fork any preceding example, wherein the bumper carrier comprises an upper portion to radially constrain the bumper.

[0075] Example 6 includes the front fork any preceding example, further including a body rigidly coupled the fourth end of the second tube, the bumper carrier is threadably coupled with the body, such that rotation of the bumper carrier causes the bumper carrier and the bumper to move axially in the second tube.

[0076] Example 7 includes the front fork of any preceding example, further including a seal head disposed in the first tube, wherein the seal head is to contact the bumper during the bottoming-out event such that the bumper is compressed between the seal head and the bumper carrier.

[0077] Example 8 includes the front fork any preceding example, wherein the body has external threads and the bumper carrier has internal threads engaged with the external threads of the body.

[0078] Example 9 includes the front fork of any preceding example, further including an adjuster pin extending through an opening in the fourth end of the second tube, the adjuster pin accessible by a user, wherein rotation of the adjuster pin causes rotation of the bumper carrier.

[0079] Example 10 includes the front fork of any preceding example, wherein the adjuster pin is rotatable to a plurality of discrete positions via a ball and detent mechanism.

[0080] Example 11 includes the front fork of any preceding example, further including a coupler, the adjuster pin and the coupler being rotationally coupled such that rotation of the adjuster pin causes rotation of the coupler.

[0081] Example 12 includes the front fork of any preceding example, wherein the coupler and the bumper carrier are rotationally coupled such that rotation of the coupler causes rotation of the bumper carrier.

[0082] Example 13 includes the front fork of any preceding example, wherein the adjuster pin extends into a bore of the coupler such that the coupler is axially slidable along the adjuster pin.

[0083] Example 14 includes the front fork of any preceding example, further including a shaft coupled to a piston in the first tube, the coupler rigidly coupled to the shaft, the coupler having a plate portion disposed between two cushioning members.

[0084] Example 15 includes the front fork of any preceding example, further including a bottom-out pad disposed on a shoulder formed in the second tube, the bottom-out pad to be engaged by the first tube when the first and second tubes are in the fully compressed position.

[0085] Example 16 is a front fork for a bicycle, the front fork comprising: a leg including an upper tube and a lower tube, the upper tube having a first top end and a first bottom end, the lower tube having a second top end and a second bottom end, the upper and lower tubes configured in a telescopic arrangement with the first bottom end of the upper tube disposed in the lower tube; a bumper disposed in the lower tube near the second bottom end; and an adjuster pin to cause axial movement of the bumper in the lower tube, wherein the bumper is moveable between a lower-most position in which a top of the bumper is a first distance from the second bottom end of the lower tube and an upper-most position in which the top of the bumper is a second distance from the second bottom end of the lower tube, the second distance being greater than the first distance.

[0086] Example 17 includes the front fork of any preceding example, further including a bumper carrier disposed in the lower tube, the bumper coupled to the bumper carrier.

[0087] Example 18 includes the front fork of any preceding example, further including a body rigidly coupled the second bottom end of the lower tube, the bumper carrier is threadably coupled with the body, wherein rotation of the adjuster pin causes rotation of the bumper carrier, which causes the bumper carrier to move axially.

[0088] Example 19 includes the front fork of any preceding example, wherein the body has external threads and the bumper carrier has internal threads engaged with the external threads of the body.

[0089] Example 20 includes the front fork of any preceding example, wherein the bumper carrier has an upper portion to radially constrain the bumper.

[0090] The illustrations of the embodiments described herein are intended to provide a general understanding of the structure of the various embodiments. The illustrations are not intended to

serve as a complete description of all of the elements and features of apparatus and systems that utilize the structures or methods described herein. Many other embodiments may be apparent to those of skill in the art upon reviewing the disclosure. Other embodiments may be utilized and derived from the disclosure, such that structural and logical substitutions and changes may be made without departing from the scope of the disclosure. Additionally, the illustrations are merely representational and may not be drawn to scale. Certain proportions within the illustrations may be exaggerated, while other proportions may be minimized. Accordingly, the disclosure and the figures are to be regarded as illustrative rather than restrictive.

[0091] While this specification contains many specifics, these should not be construed as limitations on the scope of the invention or of what may be claimed, but rather as descriptions of features specific to particular embodiments of the invention. Certain features that are described in this specification in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

[0092] Although specific embodiments have been illustrated and described herein, it should be appreciated that any subsequent arrangement designed to achieve the same or similar purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all subsequent adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, are apparent to those of skill in the art upon reviewing the description.

[0093] The Abstract of the Disclosure is provided to comply with 37 C.F.R. § 1.72(b) and is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, various features may be grouped together or described in a single embodiment for the purpose of streamlining the disclosure. This disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter may be directed to less than all of the features of any of the disclosed embodiments. Thus, the following claims are incorporated into the Detailed Description, with each claim standing on its own as defining separately claimed subject matter.

[0094] It is intended that the foregoing detailed description be regarded as illustrative rather than limiting and that it is understood that the following claims including all equivalents are intended to define the scope of the invention. The claims should not be read as limited to the described order or elements unless stated to that effect. Therefore, all embodiments that come within the scope and spirit of the following claims and equivalents thereto are claimed as the invention.

Claims

1. A front fork for a bicycle, the front fork comprising: a leg including a first tube and a second tube, the first tube having a first end and a second end opposite the first end, the second tube having a third end and a fourth end opposite the third end, the first and second tubes configured in a telescopic arrangement, the first and second tubes moveable between a fully extended position and a fully compressed position; and a bumper disposed in the second tube near the fourth end, the bumper to be contacted during a bottoming-out event before the first and second tubes reach the fully compressed positioned, wherein a position of the bumper is adjustable in the second tube.
2. The front fork of claim 1, wherein the second end of the first tube is disposed within the second tube.

3. The front fork of claim 1, wherein the second tube includes a wheel attachment portion extending from the fourth end, the wheel attachment portion to be coupled to a hub on a wheel of the bicycle.
4. The front fork of claim 1, further comprising a bumper carrier in the second tube, the bumper coupled to the bumper carrier, wherein the bumper carrier is axially moveable to adjust the position of the bumper in the second tube.
5. The front fork of claim 4, wherein an upper portion of the bumper carrier radially constrains the bumper.
6. The front fork of claim 4, further comprising a body rigidly coupled the fourth end of the second tube, the bumper carrier is threadably coupled with the body, such that rotation of the bumper carrier causes the bumper carrier and the bumper to move axially in the second tube.
7. The front fork of claim 4, further comprising a seal head disposed in the first tube, wherein the seal head is configured to contact the bumper during the bottoming-out event such that the bumper is compressed between the seal head and the bumper carrier.
8. The front fork of claim 6, wherein the body has external threads and the bumper carrier has internal threads engaged with the external threads of the body.
9. The front fork of claim 4, further comprising an adjuster pin extending through an opening in the fourth end of the second tube, the adjuster pin accessible by a user, wherein rotation of the adjuster pin causes rotation of the bumper carrier.
10. The front fork of claim 9, wherein the adjuster pin is rotatable between a plurality of discrete positions via a ball and detent mechanism.
11. The front fork of claim 9, further including a coupler, the adjuster pin and the coupler being rotationally coupled such that rotation of the adjuster pin causes rotation of the coupler.
12. The front fork of claim 11, wherein the coupler and the bumper carrier are rotationally coupled such that rotation of the coupler causes rotation of the bumper carrier.
13. The front fork of claim 12, wherein the adjuster pin extends into a bore of the coupler such that the coupler is axially slidable along the adjuster pin.
14. The front fork of claim 12, further including a shaft coupled to a piston in the first tube, the coupler rigidly coupled to the shaft, the coupler having a plate portion disposed between two cushioning members.
15. The front fork of claim 1, further including a bottom-out pad disposed on a shoulder formed in the second tube, the bottom-out pad to be engaged by the first tube when the first and second tubes are in the fully compressed position.
16. A front fork for a bicycle, the front fork comprising: a leg including an upper tube and a lower tube, the upper tube having a first top end and a first bottom end, the lower tube having a second top end and a second bottom end, the upper and lower tubes configured in a telescopic arrangement with the first bottom end of the upper tube disposed in the lower tube; a bumper disposed in the lower tube near the second bottom end; and an adjuster pin to cause axial movement of the bumper in the lower tube, wherein the bumper is moveable between a lower-most position in which a top of the bumper is a first distance from the second bottom end of the lower tube and an upper-most position in which the top of the bumper is a second distance from the second bottom end of the lower tube, the second distance being greater than the first distance.
17. The front fork of claim 16, further including a bumper carrier disposed in the lower tube, the bumper coupled to the bumper carrier.
18. The front fork of claim 17, further including a body rigidly coupled the second bottom end of the lower tube, the bumper carrier is threadably coupled with the body, wherein rotation of the adjuster pin causes rotation of the bumper carrier, which causes the bumper carrier to move axially.
19. The front fork of claim 18, wherein the body has external threads and the bumper carrier has internal threads engaged with the external threads of the body.

20. The front fork of claim 17, wherein the bumper carrier has an upper portion to radially constrain the bumper.
